

Siva Balan Saravanan

New York, NY (Open to relocation) | sivabalan212k@gmail.com | NYU Email: ss21176@nyu.edu | +1 (201) 589-8630
[linkedin.com/in/siva-balan-063581189](https://www.linkedin.com/in/siva-balan-063581189) | GitHub: github.com/sivabalan21 | Portfolio: sivabalan21.github.io

Education

New York University, Courant Institute of Mathematical Sciences

Sep 2025 – May 2027

M.S. in Computer Science | GPA: 3.67/4.0

New York, NY

- Coursework: Building LLM Reasoners, Machine Learning, GPU Programming and Architecture, Technologies for Finance, Realtime Big Data Analytics, Big Data Application Development, Fundamental Algorithms, Operating Systems, Programming Languages

Amrita School of Engineering

Jul 2018 – Jun 2022

B.Tech in Computer Science and Engineering | GPA: 8.6/10.0

India

Professional Experience

Ezee.ai - Member of Technical Staff | *Node.js, Express, MongoDB, MySQL, Redis, Kafka, AWS* Jun 2022 – Jul 2025

Chennai, India

- Built and owned **~200 REST APIs** across microservices for a debt collection platform used by **~25 financial institutions**, spanning debt collection management, performance management, and shared platform services
- Re-architected a mission-critical batch pipeline for a **9x runtime cut (180 min → 20 min)** via MongoDB aggregation tuning, compound indexes, Redis caching, field projection, and bulk writes, eliminating a long-standing performance bottleneck that had persisted for over two years.
- Designed **3 reusable platform services** used across the product: IAM-inspired **Access Management** (~50 APIs, JWT + RBAC), multi-channel **Notification** (Email/SMS/WhatsApp/Push), and multi-level **Approval Workflow** with dynamic, business rules.
- Led technical design discussions for multiple enterprise features, collaborated directly with enterprise financial institution clients to translate business requirements into scalable system designs, reviewed PRs, mentored junior engineers, and drove production releases.
- Resolved **30+ production incidents**; when a query deleted a live MongoDB collection, reconstructed the data from logs and dependent collections within **10 hours** to restore the client, then authored the backup/recovery standard adopted company-wide.

Ezee.ai - Backend Development Intern

Feb 2022 – May 2022

Chennai, India

- Delivered a reusable backend component library and generic API versioning framework adopted across future features, cutting development effort by **50%**.

Selected Projects

HealthcareAI | *Node.js, React, PostgreSQL, Redis - GitHub, Demo*

Feb 2026

- Architected a HIPAA-compliant AI SaaS spanning **13 clinical workflows** with a multi-provider AI orchestration layer (Groq, MedGemma, Ollama); **finalist among 40 teams** at Columbia's AI for Good Hackathon.
- Shipped AI-assisted medical imaging, voice-to-SOAP, and lab-extraction features backed by reusable RBAC **Access Control** and multi-channel **Notification** microservices.

E-Commerce Recommendation Platform | *Spark, Kafka, Ray*

Apr 2026 – May 2026

- Trained a collaborative-filtering recommendation model on **75M+ Amazon reviews** using Spark MLlib ALS, re-ranked with GraphFrames PageRank.
- Streamed live orders via **Kafka** and tuned the model with **Ray Tune**, deploying it as a REST API on **Ray Serve**.

Building LLM Reasoners: End-to-End LLM Systems | *Python, PyTorch, Triton* - NYU Coursework

Jan 2026 – Mar 2026

- Built a Transformer LM from scratch - BPE tokenizer, RoPE, SwiGLU, AdamW cosine schedule, reaching **≤1.45 validation cross-entropy** on TinyStories.
- Implemented **FlashAttention-2** from scratch in Triton (forward + backward, BF16), then used it to fine-tune Qwen2.5-Math-1.5B via SFT and **GRPO**, reaching **≥30% accuracy** on the Countdown reasoning benchmark.

FlashAttention-3 Engineering on H100 GPUs | *CUDA, CUTLASS, Triton*

Mar 2026 – Apr 2026

- Engineered and benchmarked **8 GPU kernels** in CUTLASS 3.x, isolating the speed impact of each H100 hardware optimization (TMA, WGMMMA, warp specialization, FP8) to pinpoint the highest-leverage techniques.
- Boosted inference throughput **2x** by adopting FP8 precision while cutting numerical error **2.6x** via block quantization, validated on Qwen2.5-7B across the GSM8K and MATH benchmarks.

Technical Skills

Languages: Python, C, C++, Java, JavaScript, SQL, CUDA

Backend: Node.js, Express.js, REST APIs, GraphQL, Microservices, JWT, RBAC

AI/ML: PyTorch, Triton, FlashAttention, Ray, Spark MLlib, RAG, Semantic Search, Hybrid Search, LangChain, LangGraph

Distributed Systems: Kafka, Spark, Hadoop, Hive, Redis, RabbitMQ, Apache ZooKeeper

Cloud, DevOps and Tools: AWS, GCP, Docker, Jenkins, Git, Jira, Postman, Tableau, Claude Code, VSCode

Databases: MongoDB, PostgreSQL, MySQL, pgvector

Achievements

Represented NYU at ACM ICPC North America Greater NY Regional (Columbia University)

Oct 2025

Won Ezee.ai AI Hackathon - Built an LLM-powered lending workflow generator using RAG

May 2025

Innovation Instigator Award, Ezee.ai - Recognized for architectural design, feature ownership, and performance optimization

2025

CodeChef 4★ (Max Rating: 1801); active competitive programmer on Codeforces and LeetCode

Present

Certifications

AWS Certified Cloud Practitioner (943/1000) | GCP Fundamentals: Services, Infrastructure, Security, ML | MongoDB: Optimizations and Performance | MTA Introduction to Programming using Python (90%)